



UofTinder

Group 80: Lance, Helen, Raghav, Nicholas, Chloe, Jessica, Milena, Vedant



Overview of Our Project

Project specification: a social networking application for users to look for their significant other, a friendship, or an academic partner

Summary of features: design will be discussed for each individual feature

- Login and logout
- Create user account and profile
- Compute compatibility scores for users
- Generate recommendation feed
- Filter recommendation feed
- Create matches between users
- Chat feature

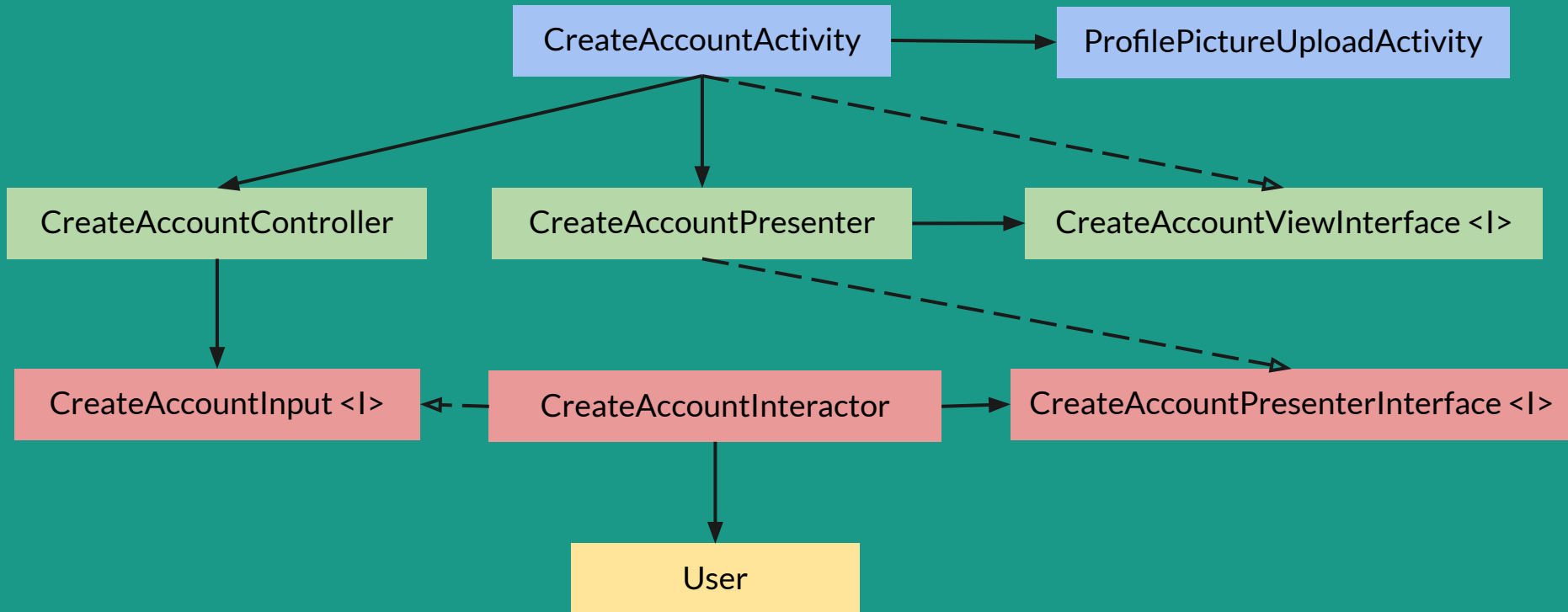


User Entity

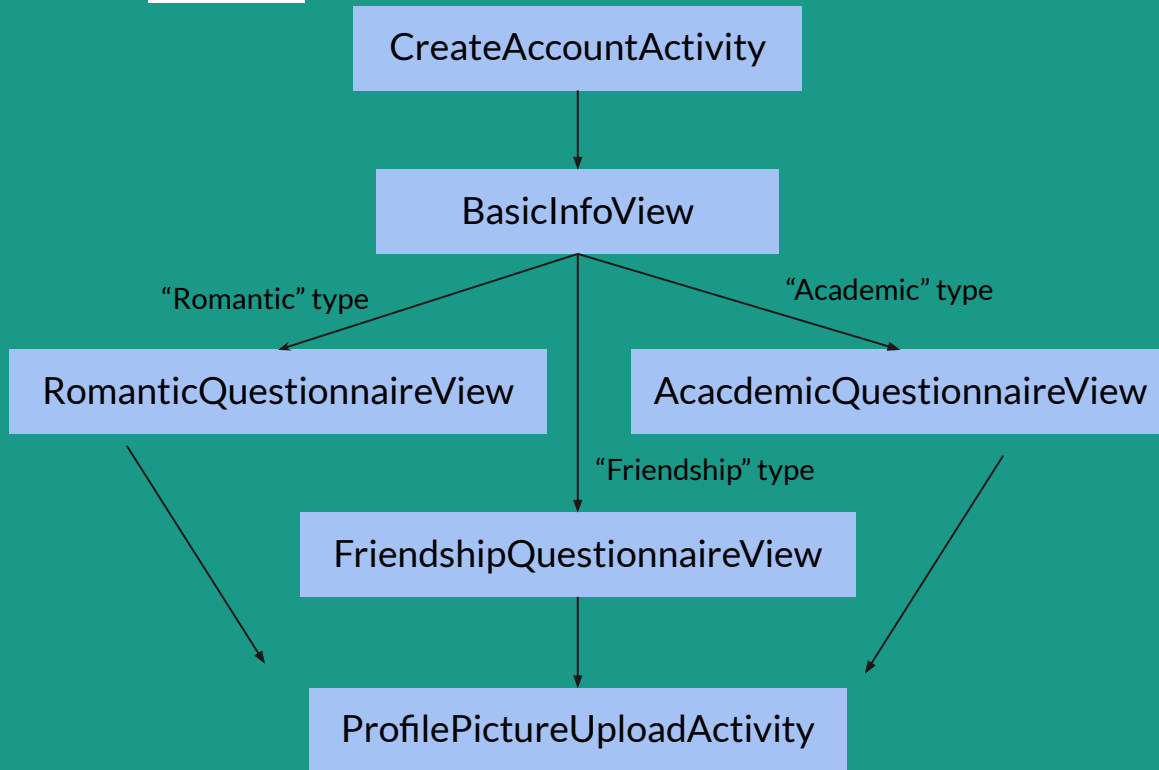
Entity shared across almost all (if not all) use cases

- `String` user ID, user type, name, gender
- `int` score, age
- `Uri` photo url
- `List<List<Integer>>` answers
- `List<String>` viewed, liked, matches

© 2006 The Authors
Journal compilation © 2006 Blackwell Publishing Ltd



Overview of UI



Login

Email:

Password:

[LOGIN](#)

Don't have an account?

[CREATE AN ACCOUNT](#)



CreateAccountActivity

- Ensures that user enters **email and password(s)**.
- Our program uses **Android Firebase** to check if the email entered is already in use.
- Ensures that user enters password twice and that password **match**
 - Password must be entered twice as it helps the user remember their password

Create An Account:

Email:

Password:

Re-enter Password:

[CREATE YOUR ACCOUNT](#)

[RETURN TO LOGIN](#)





BasicInfoView

- Ensures that user enters all **basic** information.
- **Account type determines the type of users the current user will match with**
- A different questionnaire will be brought up corresponding to the user type.

Basic Information

What is your name?

How old are you?

How do you identify?

☐ Female ☐ Male ☐ Non-binary ☐ Other/Prefer not to say

Choose your account type!

☐ Academic ☐ Friendship ☐ Romantic

CONTINUE



Academic Questionnaire

What is your year of study?

1

2

3

4

Grad

What is/are your major(s)?

Computer Science

Mathematical and Physical Science

What campus are you on?

St. George

Mississauga

Scarborough

CONTINUE

Friendship Questionnaire

What is your year of study?

1

2

3

4

Grad

What is/are your major(s)?

Computer Science

Mathematical and Physical Science

What campus are you on?

St. George

Mississauga

Scarborough

What are your interests?

Art

Fashion

Food

Gaming

Movies

What is your favourite colour(s)?

Red

Orange

Yellow

Green

Blue

CONTINUE

Romantic Questionnaire

What is your sexuality?

Heterosexual

Homosexual

Bisexual

P

What is/are your major(s)?

Computer Science

Mathematical and Physical Science

What campus are you on?

St. George

Mississauga

Scarborough

What are your interests?

Art

Fashion

Food

Gaming

Movies

Are you willing to do long distance?

Yes

No

Not Sure

What kind of relationship are you looking for?

Casual

Long Term

Don't know!

CONTINUE

ProfilePicUploadActivity

- Allows user to upload a profile picture
- Ability to skip, uploading a profile picture is optional

Upload Your Profile Picture

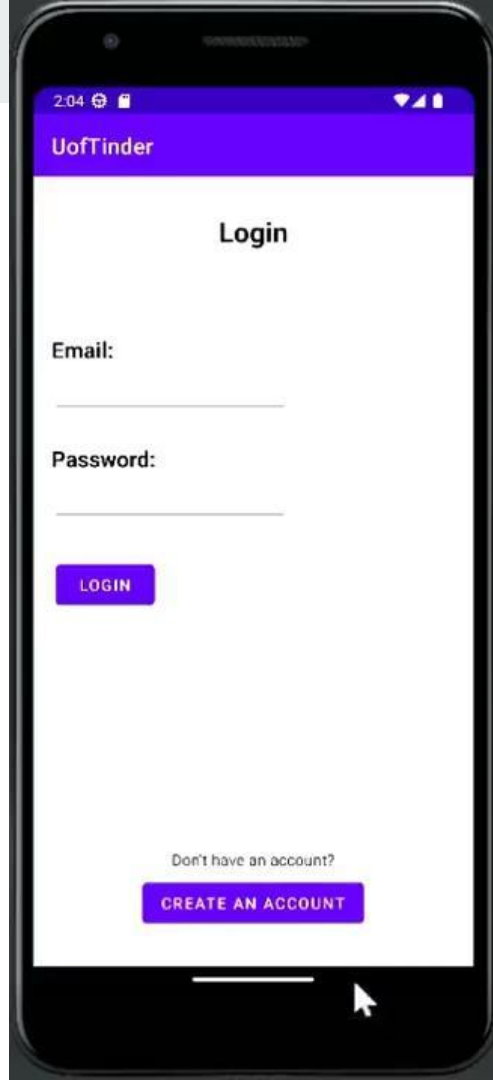


UPLOAD

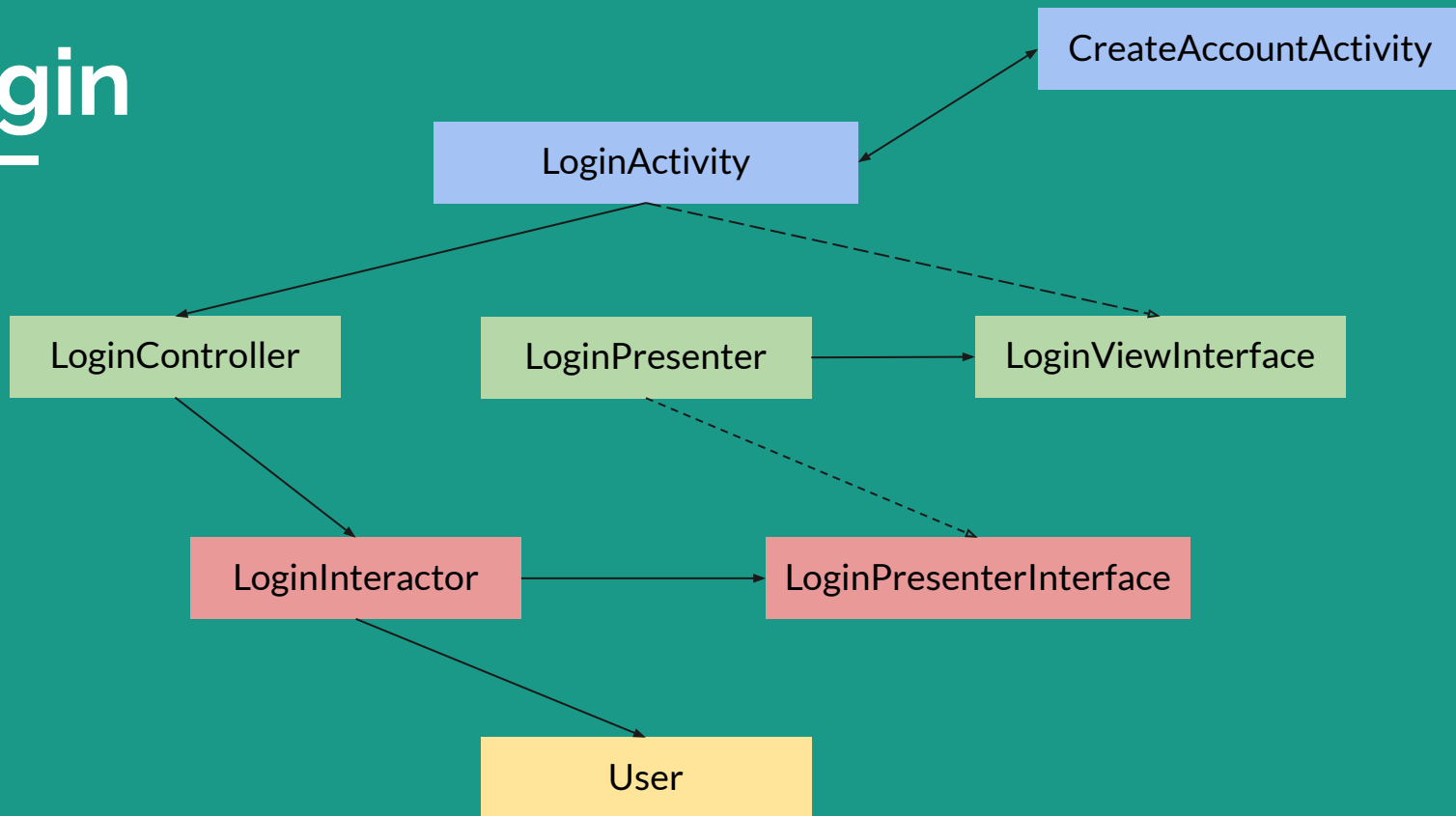
SKIP



Demo



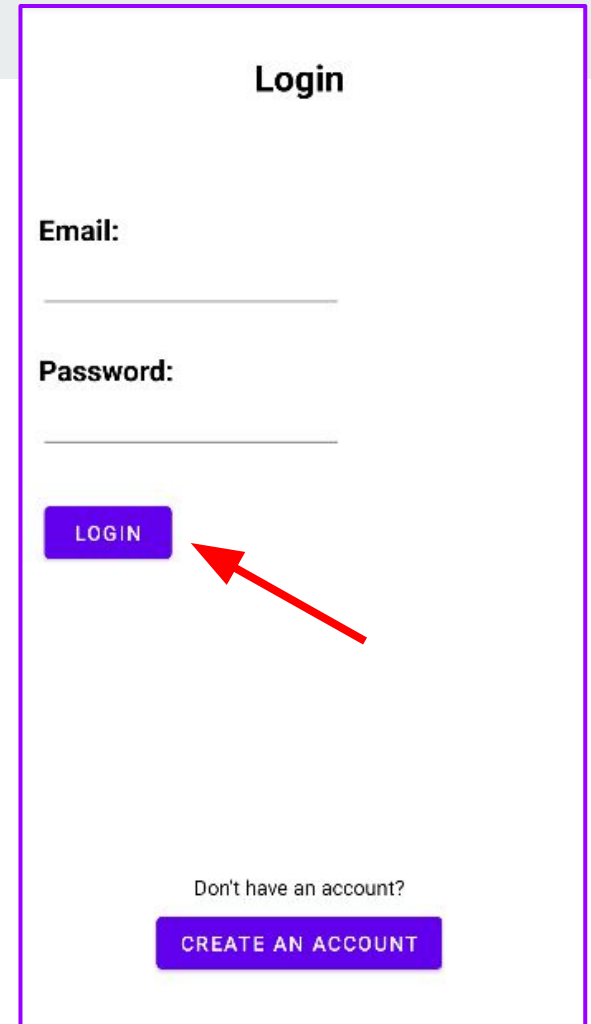
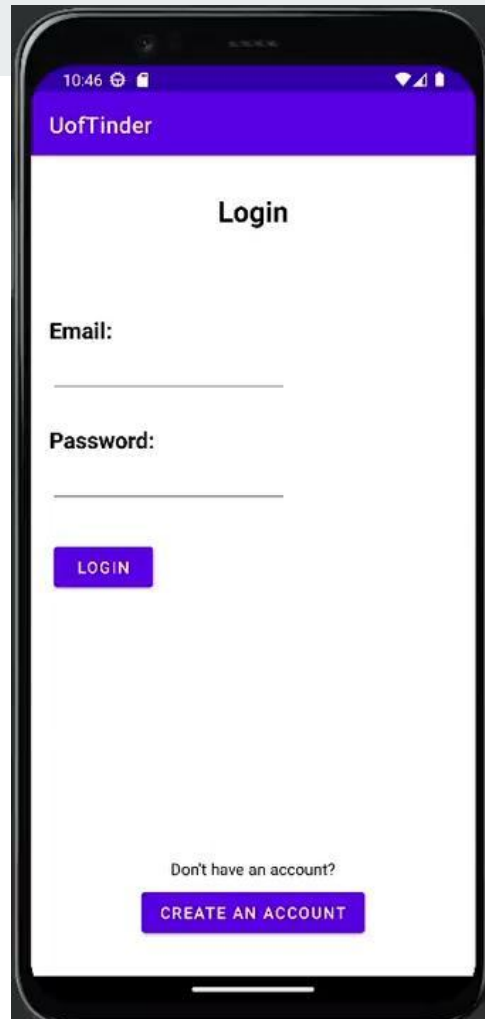
Login



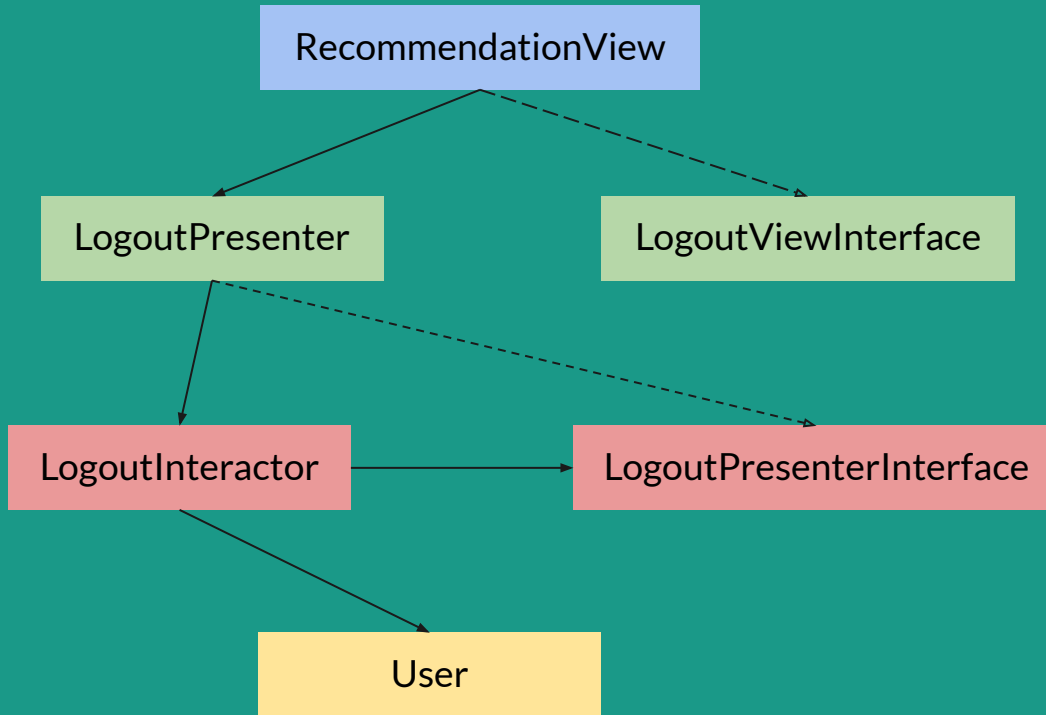
LoginActivity

- Checks if email and password inputs are not empty
- If not empty, check if a user exists in the FirebaseAuth
- Retrieves user from RealTimeDatabase according to Uid

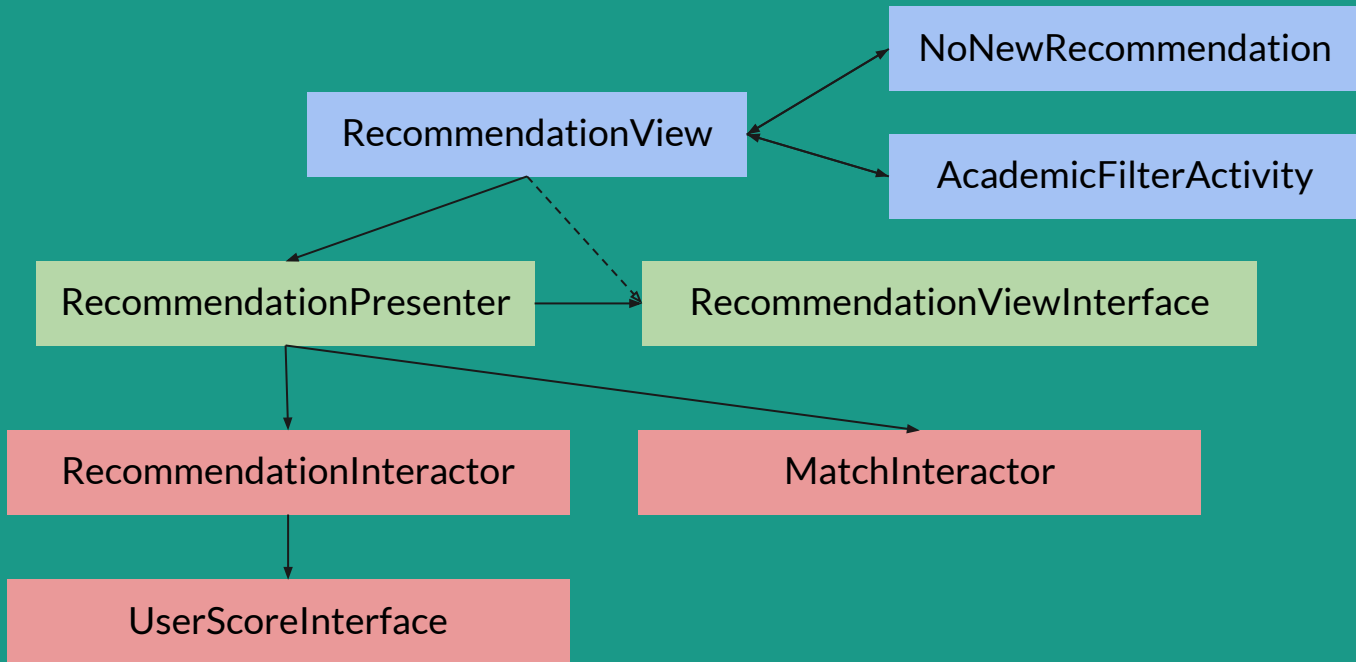
Email: milena@gmail.com
Password: 123456



Logout

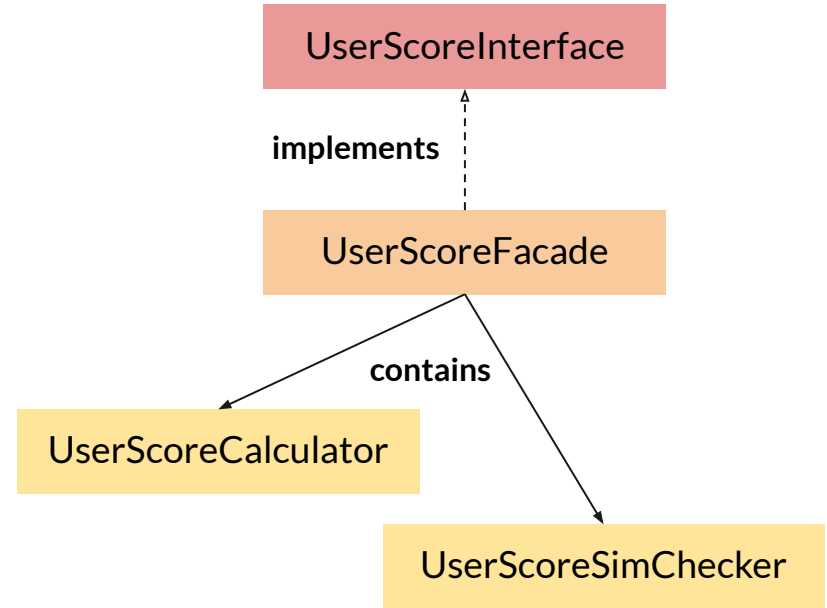


Recommendation Feed



UserScoreFacade

- Facade class that delegates tasks to calculate the current user's score and find similarity between two user scores
- Implements **UserScoreInterface**
- Contains instances of **UserScoreCalculator** and **UserScoreSimChecker**
- Uses the Facade Design Pattern!



UserScoreCalculator

Takes in the current user's answers to the questionnaire, and parses it into a series of bits!

CreateAccountInteractor getters tell us how to calculate this score:

- **answerSchema** - contains the number of multiple choice options each question has
- **isMultiSelect** - contains whether each question can only have one option selected or multiple

Questionnaire

What campus are you on?

St. George Mississauga Scarborough

What are your interests?

Art Fashion Food Gaming Movies/

Campus: St. George

Interests: Fashion,
Gaming

userAnswers

[[0], [1, 3]]

generateCompatibilityScore

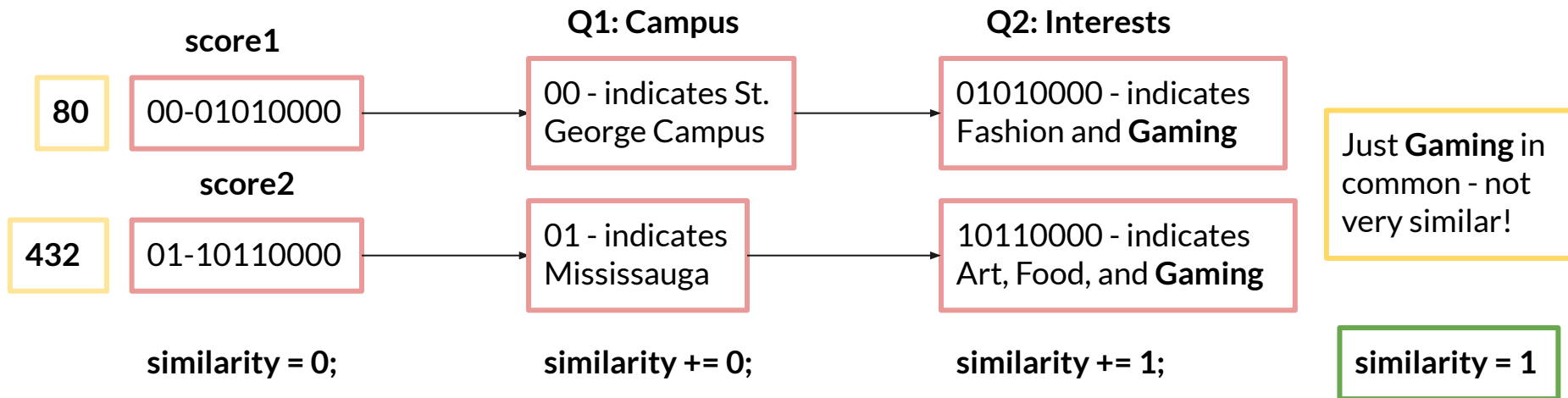
Convert to bits!

00-01010000

80

UserScoreSimChecker

Provides a way of determining how similar two users are by comparing their compatibility scores bit by bit!



RecommendationPresenter



```
graph TD; RP[RecommendationPresenter] --> RI[RecommendationInteractor]; RP --> MI[MatchInteractor];
```

RecommendationInteractor

MatchInteractor

- Model View Presenter (MVP) Design Pattern
- Called by RecommendationView
- RecommendationInteractor and MatchInteractor use cases
- Access to RecommendationView through RecommendationViewInterface



RecommendationInteractor

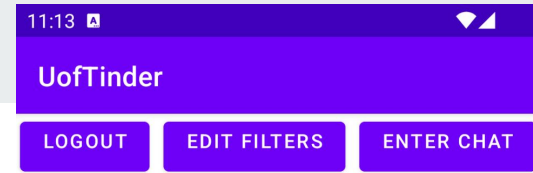
- RecommendationPresenter
- Initialization in the constructor
- compatibilityList

A Challenge I Faced During The Project

- Integrate RecommendationPresenter and RecommendationInteractor into all other functionalities of the feed

Feed: View Layer

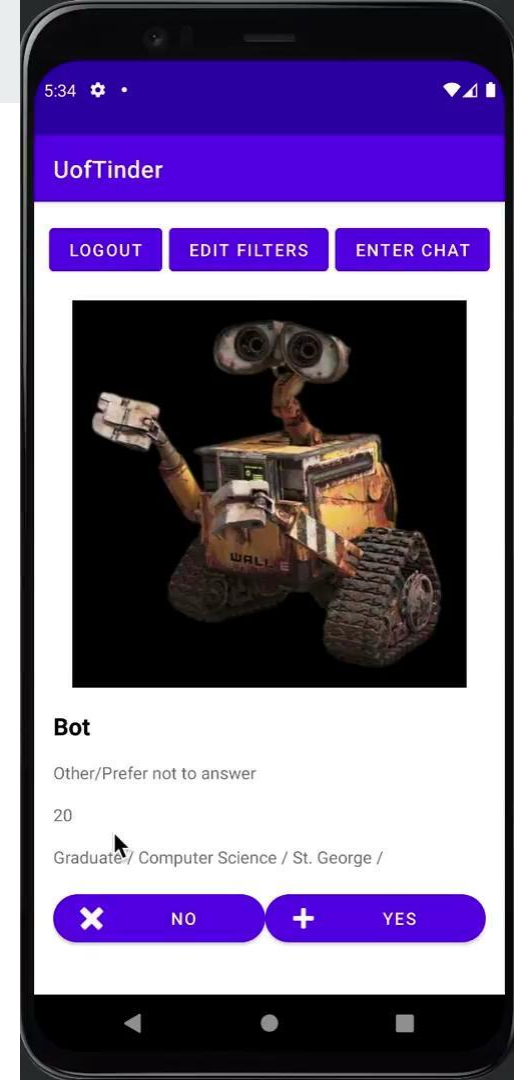
- **RecommendationViewInterface** allows the Presenter to call methods in the View to display profiles on the feed
- Connections to other Views
 - Filter feed
 - Chat
 - Logout



No more recommendations!

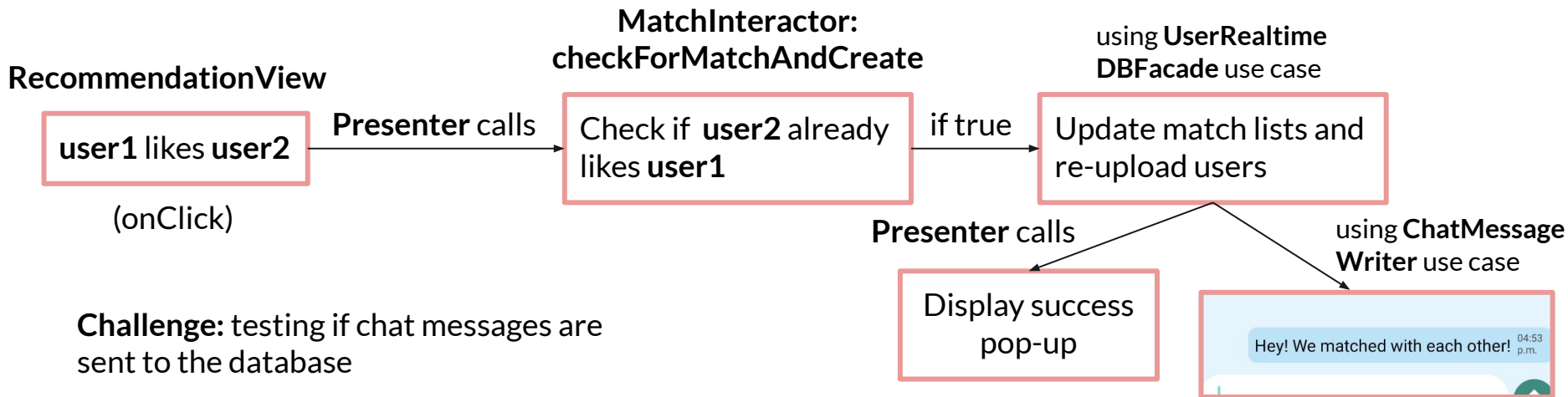
Filter Recommendation Feed

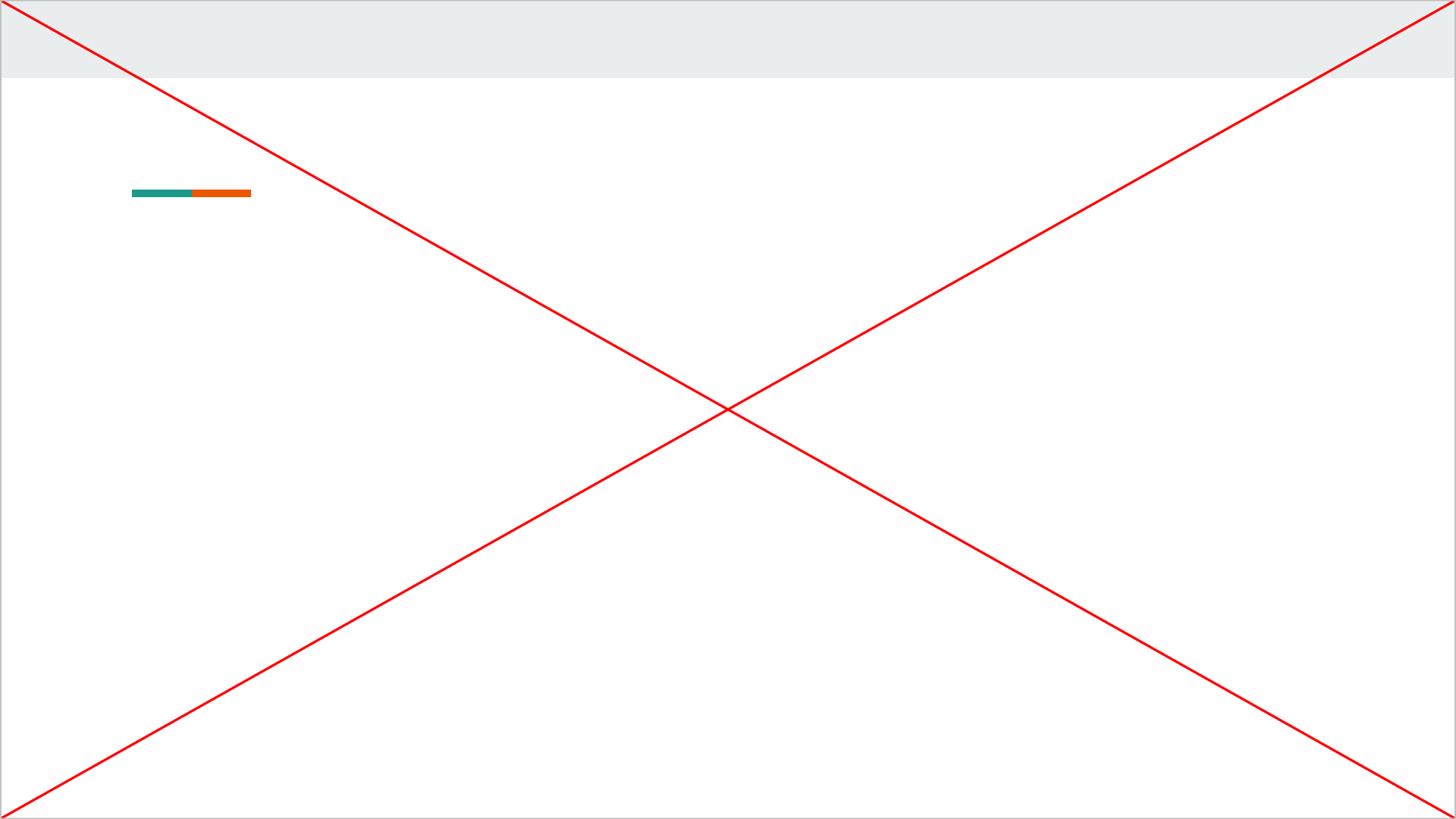
- Filters correspond to the options on the account questionnaire
- Polished aspects: age picker form validation, scrollable view
- Multi-select across several criteria groups
 - Need to satisfy one of the multi-select for a group to be satisfied
 - Need to satisfy all groups
- Design and challenges faced
 - Interactions with the generate compatibility list user story
 - Became a part of **RecommendationInteractor** to share the View and Presenter layer classes
- Future work: create UI for romantic and friendship type filters



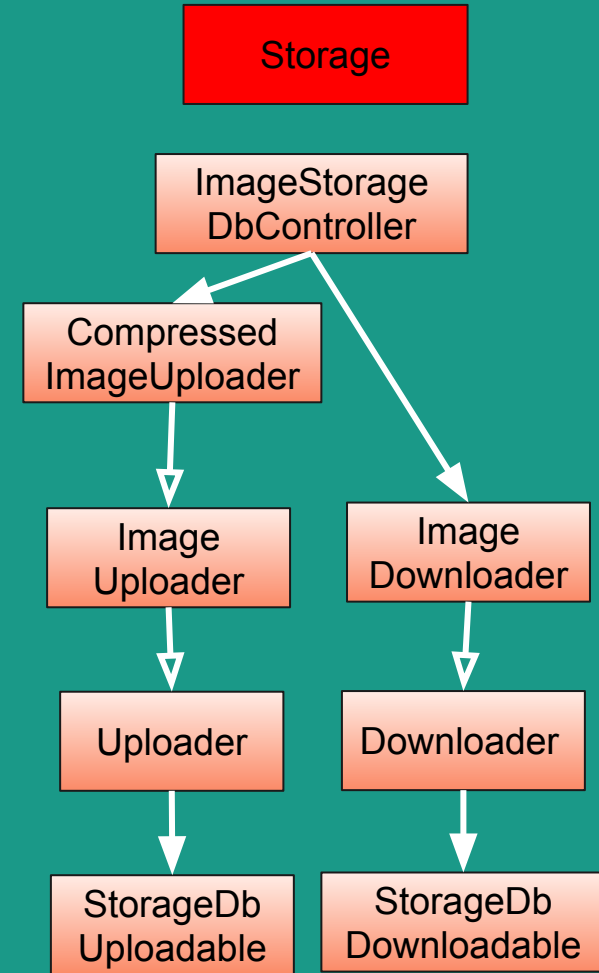
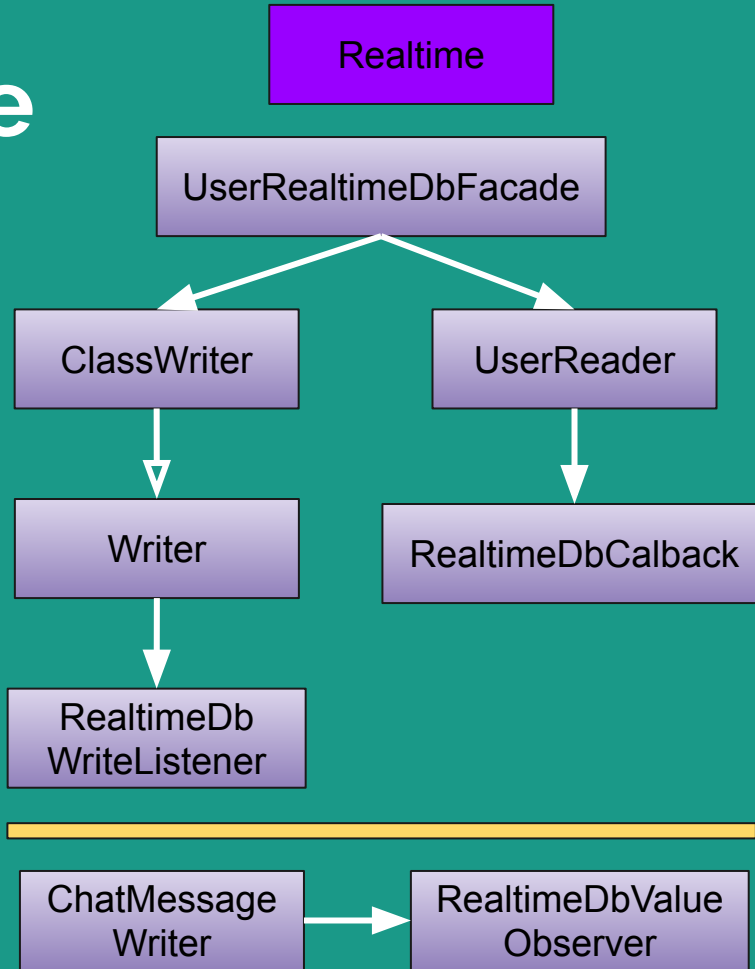
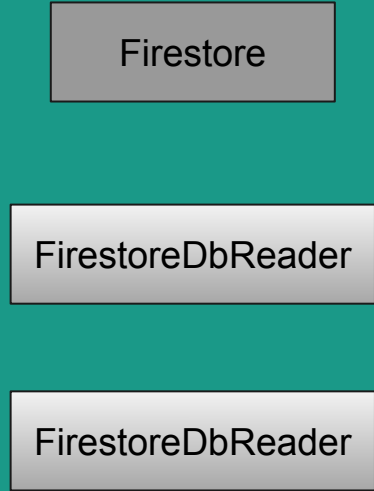
Creating a Match

Checks for match between the current user and the user they have just liked





Database



Realtime Database

Data

Rules

Backups

Usage

<https://csc207-uoftinder-default-rtdb.firebaseio.com>

<https://csc207-uoftinder-default-rtdb.firebaseio.com/>

▼ — chats

▼ — CTu0LdEN0gcPrKM3EDCITAC5KcN2qfG4IJZD0lfJmiln5FInoiS1gBy2

▼ — -NIewzS8fiHQWJwdBUUV

currentTime: "11:59 p.m."

message: "Hey! We matched with each other!"

senderId: "CTu0LdEN0gcPrKM3EDCITAC5KcN2"

timestamp: 1670389163682

▼ — -NIIf28kpB81jiXb0wFsc

<https://csc207-uoftinder-default-rtdb.firebaseio.com>

-NIIf28kpB81jiXb0wFsc + 🗑

currentTime: "12:26 a.m."

message: "Hi bot! Nice to meet you! I'm Human~"

senderId: "CTu0LdEN0gcPrKM3EDCITAC5KcN2"

timestamp: 1670390778749

<https://csc207-uoftinder-default-rtdb.firebaseio.com>

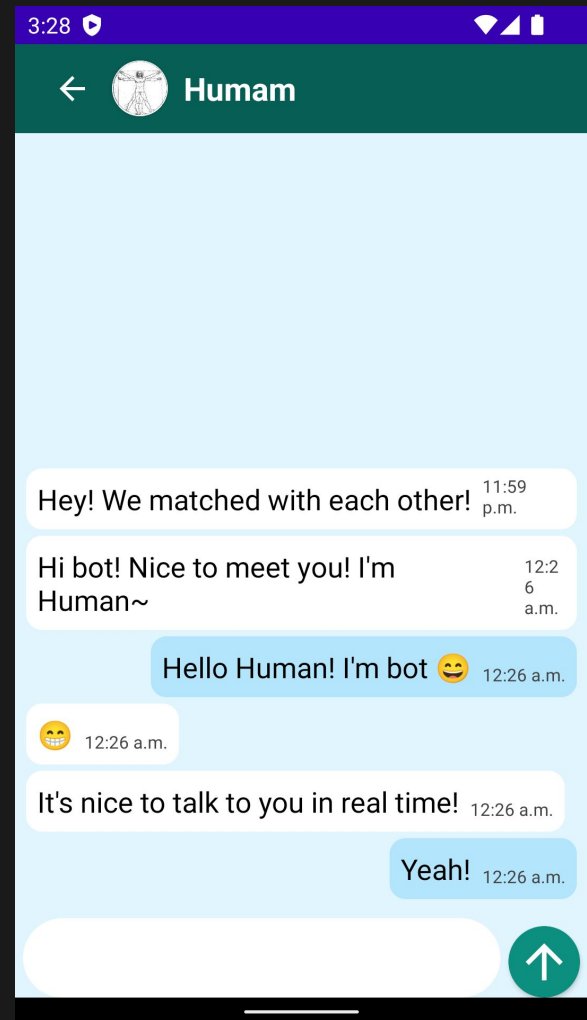
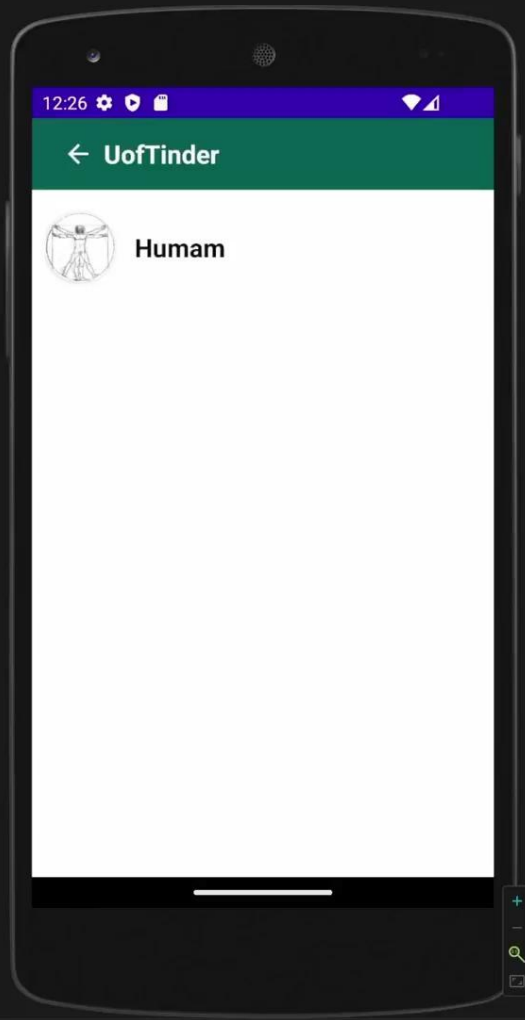
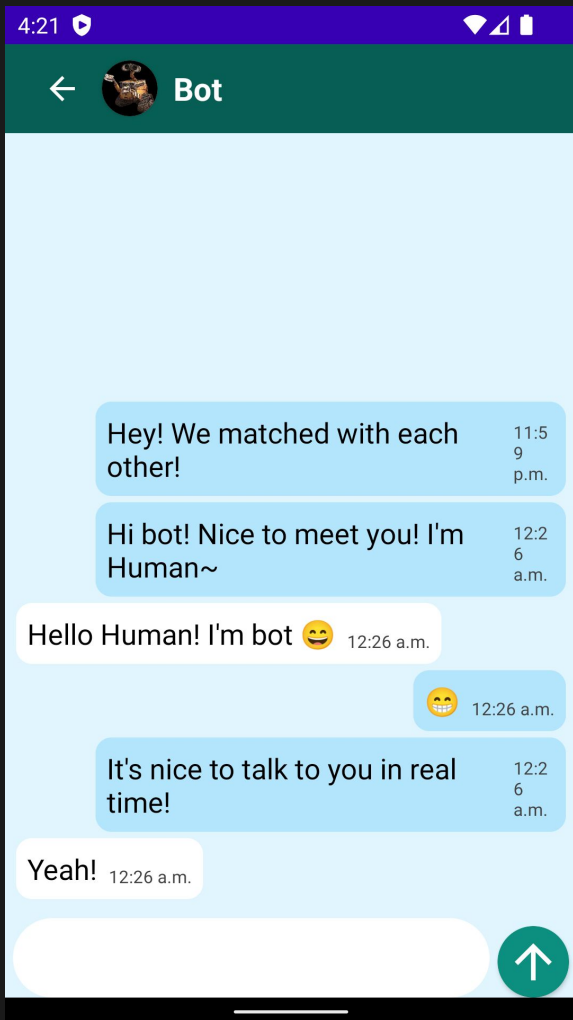
-NIIf2ChInlvCc8G9i2gc + 🗑

currentTime: "12:26 a.m." 🗑

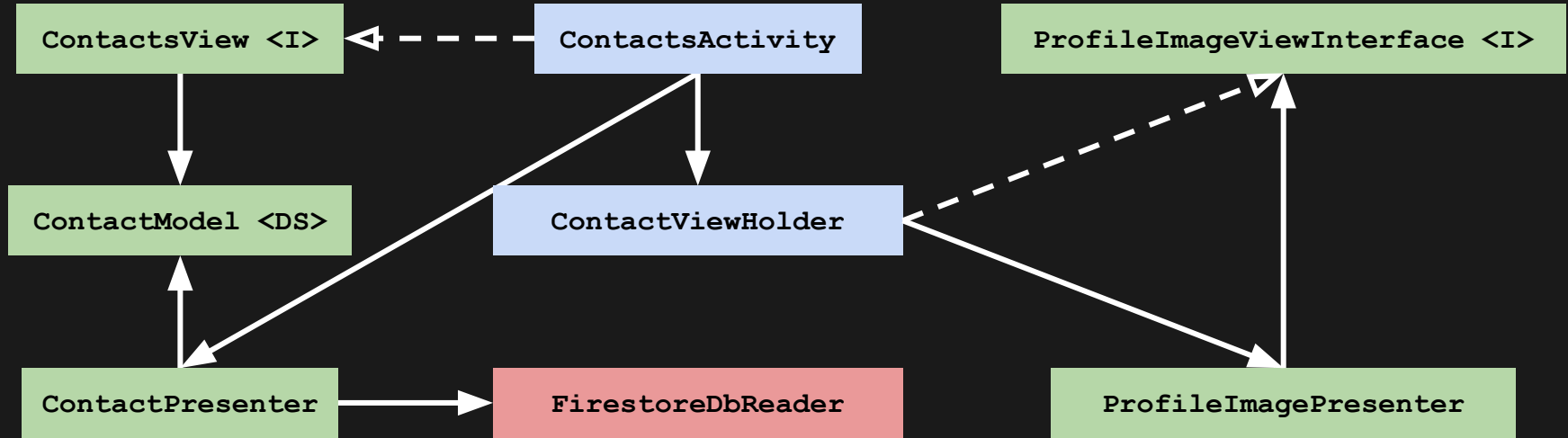
message: "Hello Human! I'm bot 😊"

senderId: "qfG4IJZD0lfJmiln5FInoiS1gBy2"

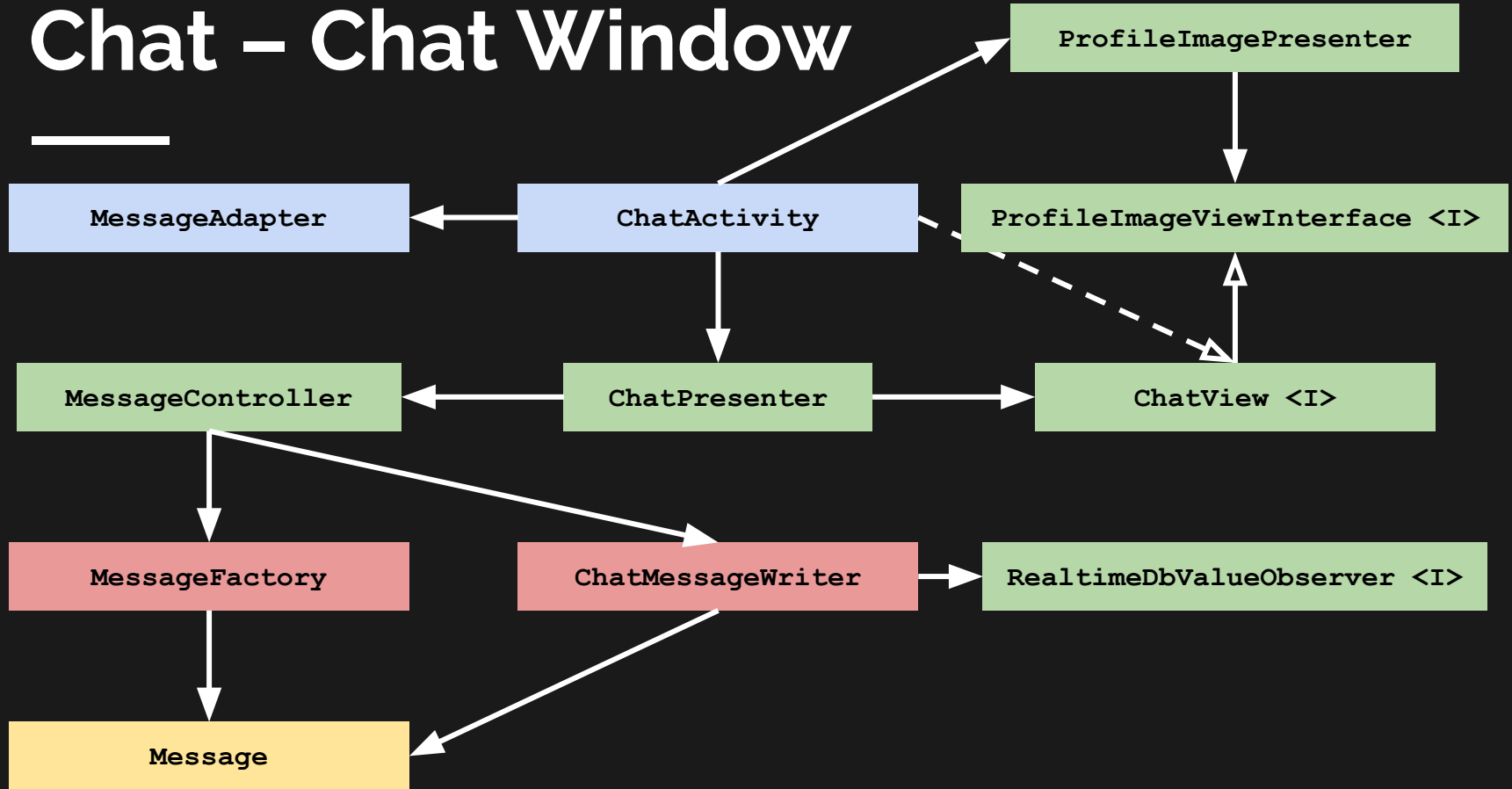
timestamp: 1670390791712



Chat – Contacts List



Chat – Chat Window



Thank you!

Happy to take any questions